

Indian Statistical Institute

BSDS - Second Year

Exercise Set 2.

1. Consider a branching process starting with a single individual, $Z_0 = 1$.
 - a) If the offspring distribution is given by $\mathbb{P}(\xi = 0) = 1/4$, $\mathbb{P}(\xi = 1) = 1/2$, $\mathbb{P}(\xi = 2) = 1/4$, compute $\mathbb{E}(Z_3)$ and $\text{Var}(Z_3)$. Also, find the probability that $\mathbb{P}(Z_3 = 0)$.
 - b) If the offspring distribution is given by $\mathbb{P}(\xi = 0) = 1/6$, $\mathbb{P}(\xi = 1) = 1/2$, $\mathbb{P}(\xi = 2) = \mathbb{P}(\xi = 3) = 1/6$, compute $\mathbb{E}(Z_2 Z_3)$ and $\text{Var}(Z_2 + Z_3)$. Also, find the probability $\mathbb{P}(Z_2 > 0, Z_3 = 0)$.
2. Suppose that $\{Z_n : n \geq 0\}$ is a branching process with $Z_0 = 1$. Fix any positive integer $r \geq 1$ and define $\{Y_n : n \geq 0\}$ as follows:

$$Y_n = Z_{nr} \text{ for } n \geq 0.$$

Check that $\{Y_n : n \geq 0\}$ is also a branching process. Find its probability generating function of the offspring distribution.

3. Consider a branching process with generating function of the offspring pgf given by $G(s) = q/(1 - ps)$ where $0 < p < 1$, $q = 1 - p$.

- a) For $p = 1/2$, show that

$$G_n(s) = \frac{(n - (n - 1)s)}{(n + 1 - ns)}$$

for $n \geq 1$. Show that $G_n(0) \rightarrow 1$ as $n \rightarrow \infty$.

- b) For $p \neq 1/2$, show that

$$G_n(s) = q \frac{p^n - q^n - (p^{n-1} - q^{n-1})ps}{p^{n+1} - q^{n+1} - (p^n - q^n)ps}.$$

Further conclude that

$$\lim_{n \rightarrow \infty} G_n(0) = \begin{cases} 1 & \text{if } p < q \\ \frac{q}{p} & \text{if } p > q. \end{cases}$$

4. Let $\{Z_n\}$ be a branching process with $Z_0 = 1$ and offspring mean μ .

a) Assume that $G'''(1) < \infty$. Show that $\mathbb{E}(Z_n Z_m) = \mu^{n-m} \mathbb{E}(Z_m^2)$ for $m \leq n$. Hence find the covariance and the correlation coefficient between Z_m and Z_n (in terms of μ) for $m \leq n$.

b) Find $\mathbb{E}(Z_n | Z_m = k)$ for $m < n$ and $k \geq 0$.

5. Find the extinction probability ρ for a branching process with offspring distribution:

a) i) $\mathbb{P}(\xi = 0) = 2/3, \mathbb{P}(\xi = 3) = 1/3$.

ii) $\mathbb{P}(\xi = 0) = 1/2, \mathbb{P}(\xi = 1) = \mathbb{P}(\xi = 3) = 1/4$.

b) Consider a branching process where each individual has a random number of offspring with a $\text{Poisson}(\lambda)$ distribution. Determine the critical value λ_c below which the branching process becomes extinct with probability 1. Find the extinction probability ρ for $\lambda > \lambda_c$.

c) Suppose that the probability generating function of the offspring distribution is given by

$$G(s) = p_0 + p_1 s + p_2 s^2$$

with $p_2 > 0, p_k = 0$ for all $k \geq 3$ and $2p_2 + p_1 > 1$. Show that the extinction probability is given by

$$\rho = \frac{p_0}{p_2}.$$

d) A branching process starts with $Z_0 = N$ individuals, each evolving independently with the same offspring distribution. Show that the probability of ultimate extinction is ρ^N , where ρ is the extinction probability starting with one individual.

6*. We set the *total progeny* of the branching process as

$$S = \sum_{n=0}^T Z_n \tag{1}$$

where $Z_0 = 1$ and T is the extinction time. We assume that $T < \infty$ with probability 1, i.e., the family becomes extinct with probability 1.

a) Show that the generating function of S is given by the solution of the following equation

$$H(s) = sG(H(s))$$

for $0 < s \leq 1$, where the probability generating function of the offspring distribution is given by $G(s)$. Show that the above equation admits a unique solution when $0 \leq s \leq 1$.

- b) Show that the total progeny of the branching process has the generating function given by

$$H(s) = \frac{1}{2ps} (1 - \sqrt{1 - 4pqs^2})$$

when the probability generating function of the offspring distribution is given by $G(s) = q + ps^2$ where $p \leq 1/2$.